Computer Vision Winter 2026 - Exercise 2

Machine Learning for Computer Vision - TU Dresden

November 2 – 6, 2026

Linear Operators

Recall the definition of linear operators $\varphi(f)(x, y)$ as seen in the lecture.

$$\varphi(f)(x,y) = \sum_{j=0}^{m_0-1} \sum_{k=0}^{m_1-1} g(j,k) f(x + j - \left\lfloor \frac{m_0 - 1}{2} \right\rfloor, y + k - \left\lfloor \frac{m_1 - 1}{2} \right\rfloor)$$

1. In Definition 12 of the lecture you have seen the 1D Gaussian average filters g_0 and g_1. Most often a discrete approximation of these filters is used in practice. Similarly, the discrete approximation of a 2D Gaussian filter is also often used:

$\frac{1}{4} \cdot$

1	2	1

Discrete 1D approximation of g_0

$\frac{1}{4} \cdot$

1
2
1

Discrete 1D approximation of g_1

$\frac{1}{16} \cdot$

1	2	1
2	4	2
1	2	1

Discrete 2D Gaussian Filter g_2

Consider a digital image f:

1	3	1
2	8	1
1	2	1

Image f

Calculate the convolution of the digital image f shown above with the discrete approximation of the 1D Gaussian filters g_0 followed by g_1, i.e. $f * g_0 * g_1$ as well as with a discrete approximation of a 2D Gaussian filter, i.e. $f * g_2$. Compare the results of $f * g_0 * g_1$ and $f * g_2$.

2. Under which condition can the convolution of a 2D filter be replaced by two (or more) convolutions with 1D filters?

3. Suppose you are given an image of size $n \times n$. How many arithmetic operations (multiplications and additions) are needed to compute the output of a convolution with a $k \times k$ dimensional 2D filter. How many arithmetic operations are required when convolving first with a $k \times 1$ dimensional filter followed by a $1 \times k$ dimensional filter. In both cases, assume the images are extended such that the final result is again an image of size $n \times n$.

4. The Sobel filter is a common choice to calculate the discrete derivative of an image in either x or y direction. The Sobel filter to calculate the x-derivative takes the following form:

$\frac{1}{8} \cdot$

1	0	-1
2	0	-2
1	0	-1

Discrete 2D Sobel Filter

What is the minimal number of arithmetic operations needed for every pixel to compute the convolution of an image with a Sobel filter?

5. Recall the operation used to sharpen an image f: $2f - (f * g_0 * g_1)$. Is it possible to rewrite this operation as a single convolution between the image f and a kernel? What would such a kernel look like?

We now want to implement the convolution of a filter with an image.

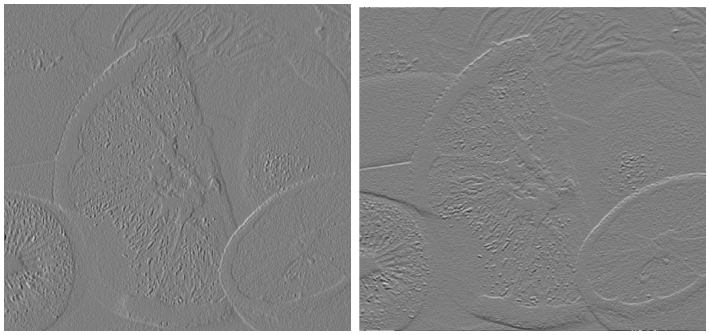
6. First we will implement a function that we will use to apply filters/operators to an image. The function to implement is the `apply_filter` function. In the `apply_filter` function loop over all pixels of the input image, apply the filter, and write the resulting pixel value to the output image as described by $\varphi(f)(x, y)$. Ensure you handle the image borders appropriately, by using zero-padding or mirror-padding such that the output image has the same size as the input image.

7. Use the `apply_filter` function implemented in task 6 to apply a Box filter g_{Box} as seen below as well as a 2D Gauss filter g_2 as seen in task 1 to the `fruits_noisy.jpg` image.

$$\frac{1}{9} \cdot \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

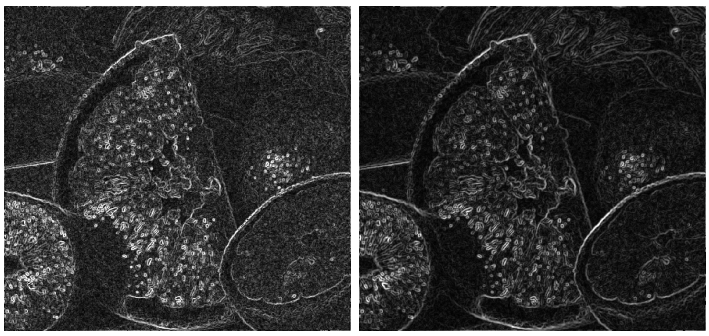
Box filter g_{Box}

8. Use the `apply_filter` function to apply Sobel filters as seen in task 4 to compute the discrete derivatives in x and y-direction.



(a) Resulting X Derivative (b) Resulting Y Derivative

9. Use the discrete derivatives computed in task 8 to compute the discrete gradient and its magnitude (i.e. the length of the gradient vector). Compute the gradient magnitude once for the `fruits_noisy.jpg` image and once for the `fruits_noisy.jpg` image after applying a Gaussian filter. Display both results.



(a) Gradient of Noisy Image (b) Gradient after Gauss Filter